Riya Mahesh

Website | LinkedIn | GitHub | riyamahesh20@gmail.com | +1 6468213164 | Google-scholar

Education

New York University, Courant Institute of Mathematical Sciences — New York
Master of Science in Computer Science, AI Concentration — ***CGPA : 3.945/4.0*** — *Sep '25 – May '27*

Indian Institute of Technology Madras — Chennai, India
Bachelor of Technology in Electrical Engineering, Minor in Computing — ***CGPA : 3.95/4.0*** — *Nov '21 - July '25*

Professional Experience (6+ months)

ML Engineering Intern, *Adobe* | *Generative AI, Inference, DiTs, PyTorch, CI/CD* — Seattle, WA
Firefly Video Generation Team — *May '26 – Aug '26*

- Deployed a **production inference API** for Adobe Firefly's video pipeline, implementing **GPU worker orchestration**, request processing, and model serving on **H100s** using **Docker, Kubernetes, and CI/CD**
- Developed a **novel inference-time** cinemagraph generation method using **WAN & LTX diffusion models**, improving motion quality by **30%**; optimized **GPU memory** for deployment; **U.S. patent application filed**

Strats and Quant Intern, *Morgan Stanley* | *Statistics, Credit Risk, Quant, Python* — Mumbai, India
Fixed Income Division — *May – Jul '24*

- Developed a **Credit & Bond Risk** pattern analysis algorithm using **feature attribution** to identify **anomaly drivers**, **deployed in production** to validate robustness during migration into new risk prediction system.

AIML Intern, *Indian Space Research Organization* | *LSTMs, Autoencoders, Stats* — Bangalore, India
Telemetry Centre — *May – Jun '23*

- Developed an **LSTM**-based anomaly detection system for satellite **time-series telemetry** with non-parametric dynamic thresholding, achieving **90% accuracy** and supporting **spacecraft health monitoring**.

R&D Experience (2+ years)

Student Researcher, CILVR Lab NYU | *VLMs, Reward Hacking, Reward Modeling* — New York
RL for Foundation Models Post-Training — *Aug '26 - Present*

- Showed partial verifiers cause **reward hacking** in RL post-training of VLMs (Qwen2.5-VL-3B, GRPO), dropping correctness **71%→38%**; trained a **reward predictor** for selective verification, cutting verifier compute by **2×**

Student Researcher, NYU Courant | *CUDA, GPU Kernels, Performance Profiling* — New York
Bit-Reproducible GPU Training for Verifiable Learning — *Sep '26 - Present*

- Built **bit-reproducible CUDA kernels** for GPT-2 training using exact integer math on **INT8 Tensor Cores**, giving **bit-identical outputs across GPUs**; profiled against cuBLAS: **2.8× faster than FP32** (int8 mode)

Research Assistant, *Nanyang Technological University* | *Distributed Systems/ ML* — Singapore
College of Computing & Data Science — *May '23 – Nov '24*

- Reduced **cross-node communication cost** in **distributed CNN training** by 30.7% using **Minimum Spanning Tree**-based **Gossip Learning** on sparse graphs (150 nodes)
- Surveyed **ML security threats** across the lifecycle, covering **Poisoning**, **Evasion**, and **Exploratory** attacks, and proposed a **taxonomy** of threat models and **cryptographic defenses**

Undergraduate Researcher, *IIT Madras* | *Sparse Attention, Causality, Time Series* — Chennai, India
Machine and Learning Theory Lab — *Aug '23 – Jul '24*

- Designed a compute-efficient **sparse attention spatio-temporal transformer** pipeline for **causal inference** of **Climate time-series data** to achieve **10%** F1/ROC improvement over **VAR/XGBoost regression** methods

Student Researcher, *NYU Langone* | *Flash Attention, DDP, Pre-Training (650M params)* — New York
Center for Human Genetics & Genomics — *Jan – May '26*

- **Pre-trained 650M-parameter ModernBERT Phylogenetic Transformer** on UniRef90 protein sequences; achieved **0.87 AUC-ROC** on ClinVar benchmark for **zero-shot** prediction of genetic variant effects

Publications / Patents

[1] Region-Constrained Video Generation (U.S. Patent filed with Adobe) (First author)
[2] Transformers with Sparse Attention for Granger Causality [Springer] (First author)
[3] Privacy and Security Implications of Cloud-Based AI Services: A Survey [arXiv] (Second author)
[4] Efficient Model Propagation for Peer-to-Peer Federated Learning using MST & Gossip Networks [ACM] (Second author)

Additional Experience

Graduate Teaching Assistant - CSCI-GA.2565-001 (7428) Machine Learning — *Spring 2026, NYU Courant*
Teaching Assistant - EE3004 Control Engineering — *Fall 2024, IIT Madras*
IEEE Student Member, IEEE Women in Engineering Member

Projects

Autonomous Rover - Software (Embedded) Systems, Anveshak — *Robotics, Centre for Innovation, IIT Madras*

- Implemented **Path Planning Algorithms - A*, Spiral search , RRT** for rover traversal
- Worked on **Aruco marker detection** with OpenCV and aruco_ros package for **3D Pose estimation**
- Established **CAN communication** between STM32 microcontrollers; implemented **PID control** for rover traversal
- Represented the team in the **International Rover Challenge - 2023**; placed **14th in the Asia-Pacific region**

Virtual Memory Management and Multithreaded Process Scheduler [GitHub] — *Operating Systems*
Built a **page-based virtual memory subsystem** and **preemptive multithreaded process scheduler**, handling context switches, address translation, and concurrent execution simulating real-time OS operations

Optimization of Traffic Signal Congestion using Reinforcement Learning — *Optimization and Control*
Implemented **Soft-Actor Critic** algorithm for signal control on simulated traffic data reducing queue length by **5%**

Pico LLM - KV Cache Transformer [GitHub] — *Deep Learning*
Developed **multi-head attention Transformer**-based LLM with **KV cache manager** for long-context generation

Information Retrieval System [GitHub] — *Natural Language Processing*
Built a custom search engine using **Vector Space Model** and **Bayesian Spell Check** & tested on Cranfield Data improving mAP, Recall, n-DCG metrics by **10%** by implementing **LSA**, **Word2Vec**, **Doc2Vec**, **BM25** algorithms

Anomaly Preserving Image Restoration for Noisy Images [GitHub] — *Deep Learning, Computer Vision*
Implemented custom **Residual Channel Attention Network**, **Denoising Autoencoders** and **U-Net** models to denoise images while preserving defects achieving highest average **PSNR** of **30.4** and **SSIM** of **99.5**

Few-shot learning using Matching, Prototypical, and Relation Networks [GitHub] — *Computer Vision*
Benchmarked three few-shot image classification models **Prototypical, Relational and Matching Networks**, reaching **98.5% accuracy** from a single labeled example per class with a Gradio interface for no-code inference

Technical Skills

Languages: Python, C++, C, Java, SQL, Unix/ Linux shell, Verilog

ML/AI: PyTorch, TensorFlow, Keras, HuggingFace Transformers, LLMs, Generative AI, Diffusion Models, Time-Series Modeling, Reinforcement Learning, LoRA, GRPO, Foundation Model Pre-training, Multimodality, Video Generation

ML Systems: Distributed Training, DDP, FlashAttention, CUDA, Mixed Precision, Model Inference, KV Caching, Parallel Programming, GPU Programming, Checkpointing, Benchmarking, CI/CD

Tools/Frameworks: Git, AWS, NumPy, Pandas, Scikit-learn, Slurm, Docker, Kubernetes, OpenCV, ROS, Claude Code, Codex, Agentic AI, Hadoop

Quantitative/Statistical: Statistical Modeling, Randomized Algorithms, Linear Optimization, Multi-Armed Bandits,Regression, Anomaly Detection, Probability, Causal Inference, Numerical Methods

Relevant Courses

Data Structures, Operating Systems, ML, Computer Organization, GPU Architecture & Programming, NLP, Control Systems, AI for Genomics, Computer Vision, Deep Learning for Imaging, Reinforcement Learning for Foundation Models, Probability, Technologies for Finance, Randomized Algorithms, Multi-Armed Bandits, Linear Optimization

Leadership

Team Head - Electrical Engineering Research Club — *IIT Madras*

- Part of 6 member EERC Core team leading 40+ members in carrying research initiatives within the EE department
- Ideated, organized and anchored events like EE Social, Professor Interviews, Scientific Talks and Blog articles

Manager - Entrepreneurship Cell, Marketing and Public Relations Division — *IIT Madras*

- Brought in **6 Public Relation deals** with prominent organizations like **The Hindu**, **Inc 42**, **Dailyhunt**
- Organised the **sustainability** campaign **Heal-Thy** at E-Summit 2023 with the theme Sustainable Space
- Organised **Stocks Are High, a virtual trading competition** during E-summit - 2022